

Young's Double slit Experiment

1. The distance between the centers of two consecutive bright fringes in Young's experiment is 0.75 mm. The distance of the screen from the slits is 68 cm. The wavelength of the light is 500nm.

- (i) Find the separation between the slits.
- (ii) Determine the distance of the 5th bright and dark fringes respectively from the central bright fringe.
- (iii) At a distance of 25 mm from the central bright fringe, determine the nature of interference (constructive or destructive).

Given

Fringe width:

$$\beta = 0.75 \text{ mm} = 0.75 \times 10^{-3} \text{ m}$$

Distance between screen and slits:

$$D = 68 \text{ cm} = 0.68 \text{ m}$$

Wavelength:

$$\lambda = 500 \text{ nm} = 5 \times 10^{-7} \text{ m}$$

Fringe width formula:

$$\beta = \frac{\lambda D}{d}$$

i. Separation between the two slits

$$d = \frac{\lambda D}{\beta}$$

$$d = \frac{(5 \times 10^{-7})(0.68)}{0.75 \times 10^{-3}}$$

$$d = 4.53 \times 10^{-4} \text{ m}$$

$$d = 0.453 \text{ mm}$$

ii. Distance of the 5th bright fringe and 5th dark fringe

Position of bright fringes:

$$y_n = n\beta$$

For the 5th bright fringe:

$$y_5 = 5\beta = 5 \times 0.75 = 3.75 \text{ mm}$$

Position of dark fringes:

$$y = \left(n - \frac{1}{2}\right) \beta$$

For the 5th dark fringe:

$$y = \left(5 - \frac{1}{2}\right) \beta$$

$$y = 4.5 \times 0.75 = 3.375 \text{ mm}$$

iii. Nature of interference at 25 mm

$$n = \frac{y}{\beta} = \frac{25}{0.75} = 33.33$$

Bright fringes occur when n is an integer and dark fringes occur when n is a half-integer.

Since 33.33 is neither an integer nor a half-integer, the point is neither bright nor dark.

Therefore, the interference is intermediate (partially constructive).

2.In Young's double-slit experiment, the distance between the centers of two consecutive bright fringes is 20 mm. The screen is placed 30 cm from the slits. The frequency of the light used is 3.0×10^{14} Hz.

- (i) Find the separation between the slits.
- (ii) Calculate the width of the dark fringes.
- (iii) What are the distances of the 3rd bright fringe and the 3rd dark fringe respectively from the central bright fringe?

Photoelectric effect Math Solution

1. A beam of ultraviolet light of wavelength 2.8×10^{-7} m is incident on a metallic surface of sodium. The stopping potential is measured to be 2.5 V.
 - (i) Calculate the maximum velocity of the emitted photoelectrons.
 - (ii) Determine the work function and the threshold frequency of the metal.
 - (iii) Investigate whether green light can produce photoelectric emission from this surface or not.
(Given: Wavelength of green light = 550 nm)

2. A light beam of wavelength 3.2×10^{-7} m is incident on a metallic plate of Nickel. The stopping potential is found to be 3.6 V.

- (i) Calculate the maximum velocity of the photoelectron.
- (ii) Calculate the work function and the threshold frequency of the metal.
- (iii) Examine whether red light can produce photoelectric current from the surface or not.
(Wavelength of red light = 730 nm)

Given:

$$\lambda = 3.2 \times 10^{-7} \text{ m}, \quad V_0 = 3.6 \text{ V}$$

Constants:

$$h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}, \quad c = 3 \times 10^8 \text{ m/s}$$

$$e = 1.6 \times 10^{-19} \text{ C}, \quad m = 9.11 \times 10^{-31} \text{ kg}$$

(i) Maximum velocity of photoelectron:

$$K_{\max} = eV_0 = (1.6 \times 10^{-19})(3.6) = 5.76 \times 10^{-19} \text{ J}$$

$$v = \sqrt{\frac{2K_{\max}}{m}} = \sqrt{\frac{2 \times 5.76 \times 10^{-19}}{9.11 \times 10^{-31}}} = \sqrt{1.264 \times 10^{12}}$$

$$\boxed{v_{\max} \approx 1.12 \times 10^6 \text{ m/s}}$$

(ii) Work function and threshold frequency:

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{3.2 \times 10^{-7}} = 9.375 \times 10^{14} \text{ Hz}$$

$$h\nu = (6.63 \times 10^{-34})(9.375 \times 10^{14}) = 6.21 \times 10^{-19} \text{ J}$$

$$\phi = h\nu - K_{\max} = 6.21 \times 10^{-19} - 5.76 \times 10^{-19} = 4.5 \times 10^{-20} \text{ J}$$

$$\boxed{\phi = 4.5 \times 10^{-20} \text{ J}}$$

$$\nu_0 = \frac{\phi}{h} = \frac{4.5 \times 10^{-20}}{6.63 \times 10^{-34}} \approx 6.8 \times 10^{13} \text{ Hz}$$

$$\boxed{\nu_0 \approx 6.8 \times 10^{13} \text{ Hz}}$$

(iii) Verification for red light:

$$\lambda_r = 730 \text{ nm} = 7.3 \times 10^{-7} \text{ m}$$

$$\nu_r = \frac{c}{\lambda_r} = \frac{3 \times 10^8}{7.3 \times 10^{-7}} = 4.11 \times 10^{14} \text{ Hz}$$

$$\nu_r > \nu_0$$

Red light will produce photoelectric current.
